Ana-Catalina Villalobos Contardo

DATA SCIENTIST & MACHINE LEARNING ENGINEER

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SUMMARY

Civil Engineer specialized in Data Science, Machine Learning, Agentic AI, and Data Engineering. Combines a rigorous mathematical foundation with production experience designing Model Context Protocol (MCP) agentic infrastructure, high-availability distributed systems, large-scale data pipelines on Google Cloud (BigQuery, Dataflow, Pub/Sub, Airflow), and predictive models. Proven track record leading multidisciplinary technical teams focused on resilient architecture and business impact.

EXPERIENCE

SimpliRoute | *Learning Engineer (Machine Learning & AI)* | August 2025 - Present

- Designed and deployed production multi-agent infrastructure using Model Context Protocol (MCP), domain subagent delegation, Human-in-the-Loop workflows, automated evaluation suites, and Redis/Kubernetes state.
- Architected a high-availability multi-provider forward/reverse geocoding platform with confidence scoring, resilient fallbacks, parallelized execution, HTTP connection pooling, and PII sanitization.
- Data modeling in Google BigQuery, distributed processing with Google Cloud Dataflow, and event orchestration with Google Pub/Sub and Apache Airflow, ensuring lineage and operational reliability.
- Automated retraining pipelines and operational monitoring for production Machine Learning predictive models.

Fracttal | *Tech Lead Fracttal Hub* | November 2023 - July 2025

- Technical leadership of an international, multidisciplinary integrations team and strategic roadmap planning for Fracttal Hub, aligning business objectives, mentoring, and continuous value delivery.
- Python development of data processing and task orchestration libraries featuring over 100 integration actions and advanced transformations across heterogeneous systems.
- Supervised and coordinated enterprise integration projects for dozens of clients, ensuring on-time delivery and high-reliability data pipelines.

Fracttal | *Data Scientist* | August 2021 - November 2023

- Designed and implemented data science processes for predictive maintenance, asset prioritization, and failure estimation.
- Collaborated on product creation and development to manage integrations between Fracttal and other platforms.

Fracttal | *Data Analyst* | February 2020 - July 2021

- Developed a predictive maintenance tool using advanced data analysis, predictive statistics, and Machine Learning techniques.
- Database design and management alongside web development tasks.

TOOLS & TECHNICAL SKILLS

- **Programming Languages:** Python (Principal/Advanced) | SQL | TypeScript | JavaScript | MATLAB.
- **Cloud & DevOps:** Google Cloud Platform (BigQuery, Dataflow, Cloud Run, Artifact Registry, Google Pub/Sub, Cloud Storage, GKE) | ArgoCD | Git | GitHub | Bitbucket.
- **Frameworks & Libraries:** FastAPI | LangChain | Scikit-learn | TensorFlow | Pandas | NumPy | Pydantic | Django | React JS.
- **Databases:** PostgreSQL | Redis | BigQuery | SQL Server | MySQL | MongoDB | SQLite.
- **Containers & Orchestration:** Airflow | Airbyte | Docker | Kubernetes.
- **Methodologies & Management:** System Design | Test-Driven Development (TDD) | Agile Methodologies (Scrum & Kanban) | Roadmap Planning | Jira | Notion.
- **Key Skills:** Agentic AI & Multi-Agent Systems (MCP, LLMs) | Distributed Systems & Geocoding | Data Pipeline Architecture (ETL/ELT) | ML Model Monitoring | Relational DB Design | Technical Leadership.

EDUCATION

- **Universidad de Chile** | *Civil Engineering with a minor in structures, construction, and geotechnics* | Graduated 2019, Degree awarded 2020 (Maximum Distinction)
  Thesis: Gaussian Processes and vibration modes for delamination detection in composite material plates.
- **Universidad de Chile** | *Bachelor of Engineering Sciences, Civil minor* | Completed 2017
  Notable activity: Teaching Assistant for Probabilities and Statistics.

PUBLICATIONS

- **Generalized Gaussian smoothing for baseline-free debonding assessment of sandwich panels** | *Structural Control Health Monitoring* - March 2021 (https://doi.org/10.1002/stc.2727)
  Methodological proposal for delamination detection and evaluation in composite plates using Gaussian Processes and Bayesian analysis.

LANGUAGES & CERTIFICATIONS

- **Languages:** Spanish (Native) | English (Advanced - EFSET English Certificate C2 Proficient, 2020)
- **Certifications:** React & Django Full Stack (Udemy, 2022) | Mastering Data Visualization (Udemy, 2022) | Applied Machine Learning in Python (Coursera, 2020).